Comparative Analysis of Research Papers on Fraud Detection, Phishing, and Mobile Financial Security

No.	Paper Name	Authors	Main Concept	Importance	Coverage Area	Limitations
1	Telecommunication Fraud Resilient Framework for Efficient and Accurate Detection of SMS Phishing Using Artificial Intelligence Techniques	Devendra Sambhaji Hapase, Lalit Vasantrao Patil	Proposed an Enhanced CNN with PCC-PCA feature extraction for SMS phishing detection in telecommunication systems.	Improves SMS phishing detection accuracy and reduces computational complexity in telecom fraud systems.	SMS phishing detection, telecom fraud, AI-based fraud classification, feature extraction, CNN models.	Focused mainly on SMS dataset real-world deployment scalability and multilingual SM handling were not deeply explored.
2	AI-Driven Cybersecurity in Mobile Financial Services: Enhancing Fraud Detection and Privacy in Emerging Markets	Ebrahim Mollik, Faisal Majeed	Investigated AI-driven fraud detection and privacy enhancement in mobile financial services (MFS).	Highlights the role of AI in strengthening cybersecurity and improving user trust in emerging fintech markets.	Mobile financial services, privacy, AI fraud detection, user trust, explainable AI, cybersecurity.	Relies heavily o survey-based perception analysis rather than implementatio of technical fraud detection systems.
3	An Integrated Fraud Defense Framework for Card-Based Account Funding in Bangladesh’s Mobile Financial Services	Sujit Kumar Sarker	Proposed a socio-technical fraud defense framework integrating Fraud Triangle Theory, Routine Activity Theory, and NIST CSF.	Important for Bangladesh’s MFS ecosystem against OTP bypass and social engineering fraud.	OTP bypass fraud, MFS fraud, cybersecurity frameworks, social engineering, Bangladesh fintech security.	Conceptual framework only lacks implementatio and experimental validation.
4	A Resilient Cybersecurity Framework for Mobile Financial Services (MFS)	Stephen Ambore, Christopher Richardson, Huseyin Dogan, Edward Apeh, David Osselton	Proposed a resilient cybersecurity framework for securing mobile financial services ecosystems.	Addresses growing cybercrime risks in mobile banking and financial inclusion platforms.	Mobile financial services, cybersecurity frameworks, human factors, cybercrime mitigation.	Preliminary framework with limited experimental validation and implementatio details.

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5	Machine Learning Approaches for Fraud Detection and Risk Assessment in Mobile Banking Applications and Fintech Solutions	Martina Ononiwu, Tony Isioma Azonuche, Onum Friday Okoh, Joy Onma Enyejo	Explored machine learning models for fraud detection and risk assessment in fintech applications.	Demonstrates how ML improves real-time fraud analysis and anomaly detection in digital finance.	Mobile banking security, fintech fraud detection, risk assessment, anomaly detection, machine learning.	Mostly review-oriented; lacks practical deployment an benchmarking experiments.
6	Design and Deployment of a Secure Mobile Banking System with AI-Driven Fraud Detection	Anh Ngo, Sindor Sapaev, Feruzbek Jumaniyozov, Zokir Mamadiyarov, Amit Mittal, Ilhom Jurayev	Developed a secure mobile banking system integrating biometrics and AI-based fraud detection.	Demonstrated very high fraud detection accuracy with improved user satisfaction.	AI fraud detection, biometric authentication, anomaly detection, secure mobile banking systems.	Prototype-focused; limite discussion on scalability and adversarial attacks.
7	Fraud Prediction and Prevention in Mobile Money Payment Systems (MMPS): A Systematic Literature Review of Text-Based Detection Methods	Umar Sayibu, Michael Asante, Gaddaf Abdul-Salam, Twum Frimpong	Systematic review of text-based fraud detection methods in mobile money systems.	Identifies research gaps and highlights the lack of NLP-based fraud detection in MMPS.	Mobile money fraud, SMS fraud, phishing, fraud prevention, machine learning, systematic review.	No original frau detection mode proposed; focused only o literature synthesis.
8	Vishing Shield: A Conceptual Framework and Detection Model to Mitigate Vishing Attacks in Mobile Financial Services (MFS)	Researchers from MIST and MBSTU Bangladesh	Proposed a conceptual vishing detection model for mitigating voice phishing attacks in MFS.	Important for addressing voice phishing threats in rapidly growing mobile financial ecosystems.	Vishing attacks, social engineering, MFS security, user awareness, fraud detection models.	Mainly conceptual and survey-based without real-world implementatio

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9	Hybrid Machine Learning Models for Adaptive Fraud Prevention in Mobile Banking Platforms	Adams Smith Williams	Proposed hybrid ML architectures combining supervised, unsupervised, and reinforcement learning.	Shows hybrid models improve adaptability and resilience against evolving fraud patterns.	Adaptive fraud prevention, ensemble learning, anomaly detection, reinforcement learning.	Theoretical analysis with limited practica experimentatio and dataset diversity.
10	A Systematic Review of AI-Enabled Fraud Detection in Digital Financial Systems (2019–2026)	Md. Rashed Buiya	Comprehensive review comparing ML, DL, hybrid, ensemble, and graph-based fraud detection approaches.	Provides quantitative evidence of the effectiveness of hybrid and ensemble AI models.	Digital finance fraud detection, AI, deep learning, graph models, ensemble learning.	Secondary research only; r original fraud detection framework implementatio
11	Machine Learning and Neural Networks for Phishing Detection: A Systematic Review (2017–2024)	Jacek Lukasz Wilk-Jakubowski, Lukasz Pawlik, Grzegorz Wilk-Jakubowski, Aleksandra Sikora	Reviewed ML and neural network approaches for phishing detection across multiple channels.	Valuable for understanding phishing detection evolution and identifying future research directions.	Website phishing, email phishing, malware, social media phishing, ML, deep learning.	Review-based research withou implementatio of a new detection mode
12	Phishing Website Detection Using Deep Learning Models	Ume Zara, Kashif Ayub, Hikmat Ullah Khan, Ali Daud, Tariq Alsahfi, Saima Gulzar	Applied ML, ensemble learning, PCA, and deep learning for phishing website detection.	Achieved very high phishing detection accuracy and improved cybersecurity reliability.	Website phishing detection, deep learning, ensemble learning, feature selection.	Mainly dataset-driven evaluation; real time deployme challenges not deeply discussed.
13	Web-Based Phishing URL Detection Model Using Deep Learning Optimization Techniques	Kousik Barik, Sanjay Misra, Raghini Mohan	Proposed EGSO-CNN optimized deep learning model for phishing URL detection.	Improves phishing detection performance with optimized feature extraction and tuning.	URL phishing detection, optimization techniques, deep learning, feature engineering.	Requires high computational resources and depends heavil on dataset quality.

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14	AI in Phishing Detection: A Bibliometric Review	Daniela Popescul, Laura Diana Radu	Bibliometric review of AI and ML applications in phishing detection research.	Helps identify research trends, influential studies, and thematic evolution in phishing detection.	AI phishing detection, bibliometric analysis, ML, deep learning, NLP.	Does not provi practical phishing detection implementatio or experimentatio
15	Detecting Voice Phishing with Precision: Fine-Tuning Small Language Models	Ju Yong Sim, Seong Hwan Kim	Developed a voice phishing detector using fine-tuned Llama3-based small language models.	Demonstrates that lightweight LLMs can effectively detect voice phishing with expert prompts.	Voice phishing detection, NLP, small language models, fine-tuning, LLM security.	Dependent on transcript quali and language-specific datase
16	Phishing Detection in the Gen-AI Era: Quantized LLMs vs Classical Models	Jikesh Thapa, Gurrehmat Chahal, Șerban Voinea Gabreanu, Yazan Otoum	Compared ML, DL, and quantized LLMs for phishing detection efficiency and robustness.	Important for understanding the future role of lightweight LLMs in phishing defense systems.	Phishing detection, GenAI, quantized LLMs, adversarial robustness, explainable AI.	LLMs still underperform compared to traditional ML/ DL models in accuracy.
17	A Hybrid Super Learner Ensemble for Phishing Detection on Mobile Devices	Routhu Srinivasa Rao, Cheemaladinne Kondaiah, Alwyn Roshan Pais, Bumshik Lee	Proposed Phish-Jam, a hybrid ensemble-based mobile phishing detection application.	Provides fast and accurate phishing detection optimized for mobile devices.	Mobile phishing detection, ensemble learning, transformer embeddings, mobile cybersecurity.	Mobile resourc constraints and dependence or URL-based features may limit broader adaptability.

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18	Deep Isolation Forest for Anomaly Detection	Hongzuo Xu, Guansong Pang, Yijie Wang, Yongjun Wang	Proposed Deep Isolation Forest (DIF) combining isolation forests with deep neural random representations for anomaly detection.	Significantly improves anomaly detection accuracy in high-dimensional and non-linear datasets while preserving scalability.	Anomaly detection, isolation forest, deep learning, ensemble learning, graph and time-series anomaly detection.	Requires computational resources for deep representation generation and may be comple for real-time deployment.
19	Anomaly Detection based on Isolation Forest and Local Outlier Factor	Abubakar Mastour Adam Fadul	Compared Isolation Forest (IForest) and Local Outlier Factor (LOF) algorithms across multiple datasets.	Provides practical comparison of anomaly detection methods using MCC, AUC, and execution time metrics.	Anomaly detection, Isolation Forest, LOF, unsupervised learning, outlier detection, fraud detection.	Focused only o two algorithms and limited benchmark datasets; lacks hybrid or deep learning approaches.
20	Comparative Analysis of Research Papers on Fraud Detection, Phishing, and Mobile Financial Security	Compiled Research Summary	Comparative review summarizing fraud detection, phishing detection, AI-based cybersecurity, and mobile financial security research papers.	Helps identify major trends, limitations, and future research opportunities in fraud and phishing detection.	Fraud detection, phishing detection, AI cybersecurity, mobile financial security, literature comparison.	Secondary summary document without experimental implementatio or original framework contribution.

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Security | Compiled Research Summary | Comparative review summarizing fraud detection, phishing detection, AI-based cybersecurity, and mobile financial security research papers. | Helps identify major trends, limitations, and future research opportunities in fraud and phishing detection. | Fraud detection, phishing detection, AI cybersecurity, mobile financial security, literature comparison. | Secondary summary document without experimental implementation or original framework contribution. |

Overall Research Trends Observed

- AI, Machine Learning, Deep Learning, and Hybrid Models are the dominant approaches for fraud and phishing detection.
- Mobile Financial Services (MFS) security is a major research focus, especially in emerging markets like Bangladesh.
- Hybrid and ensemble models generally outperform single-model systems in fraud detection accuracy.
- Social engineering attacks such as phishing, vishing, and OTP bypass are increasing concerns.
- Explainable AI (XAI), lightweight LLMs, and adaptive learning are emerging future directions.
- Many papers focus on conceptual frameworks or systematic reviews rather than real-world deployment.
- Real-time scalability, adversarial robustness, multilingual support, and privacy preservation remain major limitations across studies.